

Final Project Exploratory Data Analysis

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2025-11-20

Packages

```
library(fpp3) # Time Series Plots and Forecasting
library(corrplot) # Correlation Plot
library(forecast) # BoxCox Transformation
library(scales) # Log Scale Tick Lables
set.seed(280) # Reproducibility
```

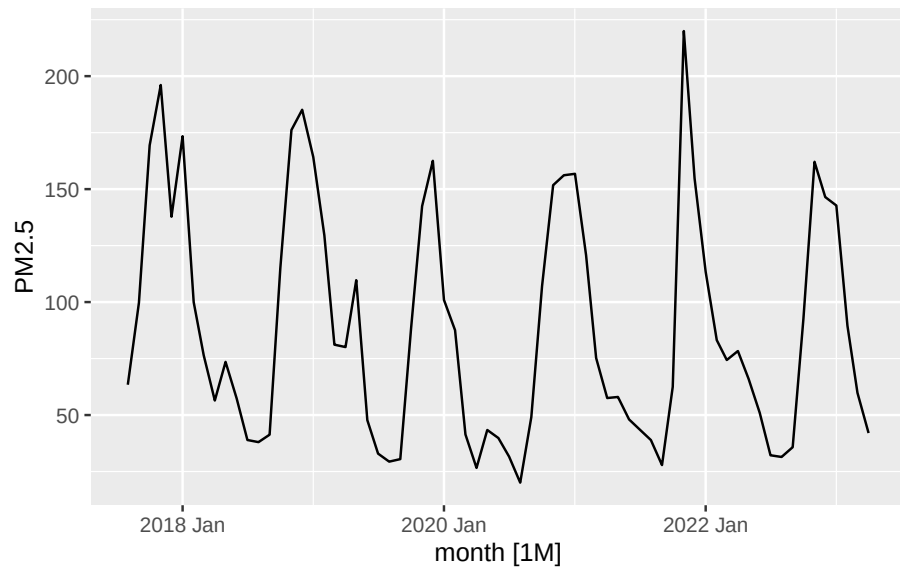
Setup

Read Data

```
data <- readRDS("../data/loaded/delhi_filter.rds")
df <- data

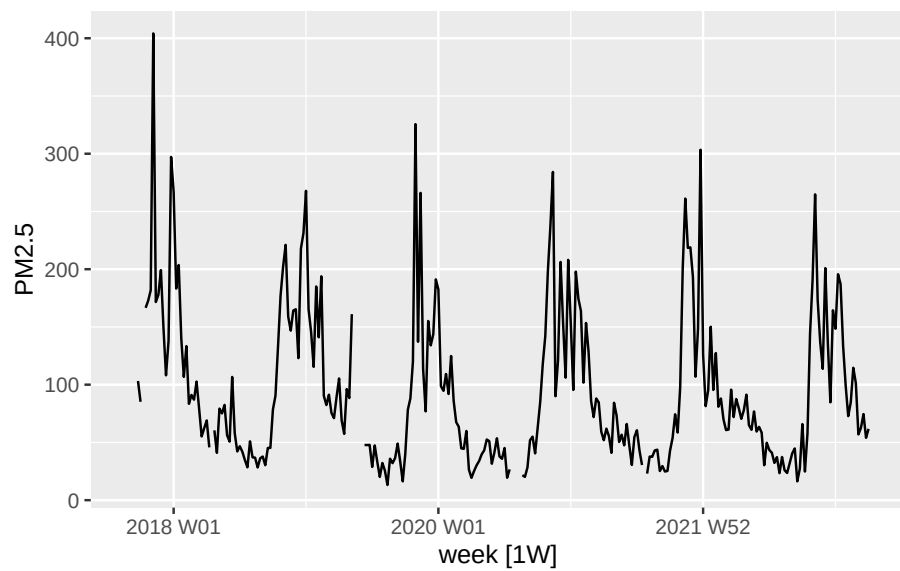
df_monthly <- data %>%
  select(PM2.5) %>%
  index_by(month = yearmonth(datetime)) %>%
  summarise(PM2.5 = median(PM2.5, na.rm = TRUE))

df_monthly %>%
  autoplot(PM2.5)
```



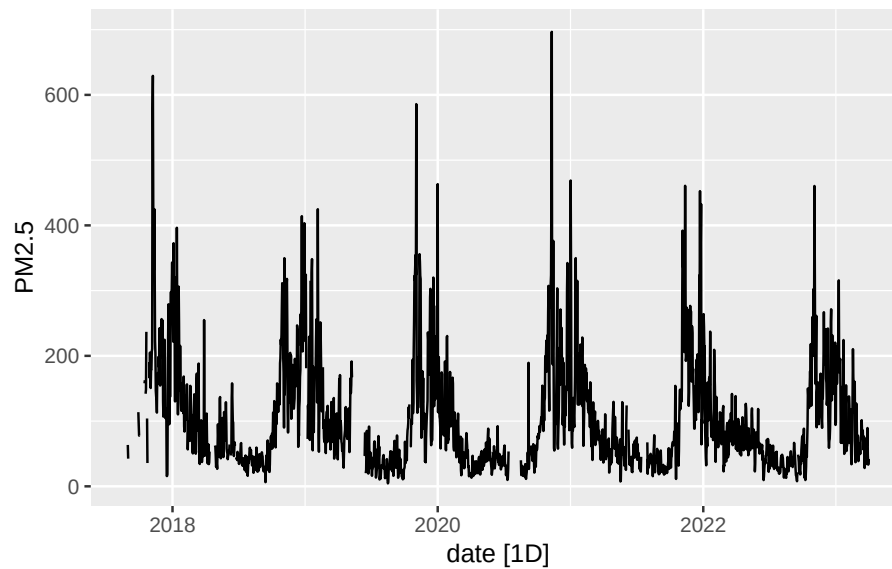
```
df_weekly <- data %>%
  select(PM2.5) %>%
  index_by(week = yearweek(datetime)) %>%
  summarise(PM2.5 = median(PM2.5, na.rm = TRUE))

df_weekly %>%
  autoplot(PM2.5)
```



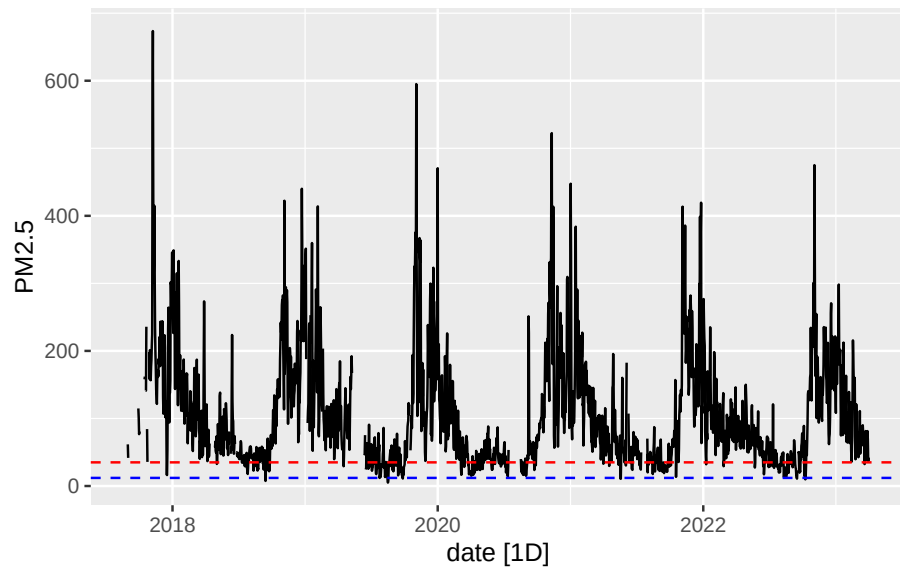
```
df_daily <- data %>%
  select(PM2.5) %>%
  index_by(date = as.Date(datetime)) %>%
  summarise(PM2.5 = median(PM2.5, na.rm = TRUE))

df_daily %>%
  autoplot(PM2.5)
```



Recomended Limit for PM2.5

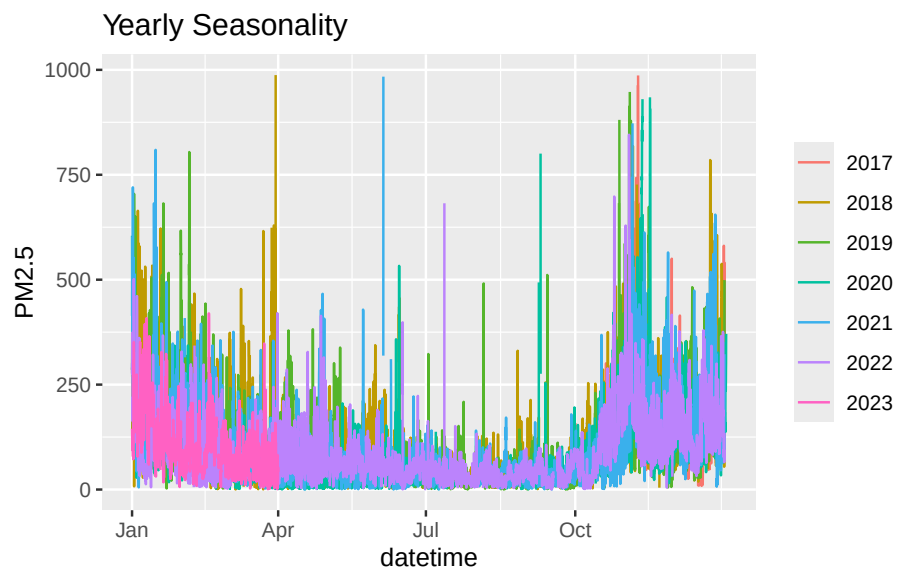
```
df %>%
  index_by(date = as.Date(datetime)) %>%
  summarise(PM2.5 = mean(PM2.5, na.rm = TRUE)) %>%
  autoplot(PM2.5) +
  geom_hline(yintercept = 35, color = "red", linetype = "dashed") +
  geom_hline(yintercept = 12, color = "blue", linetype = "dashed")
```



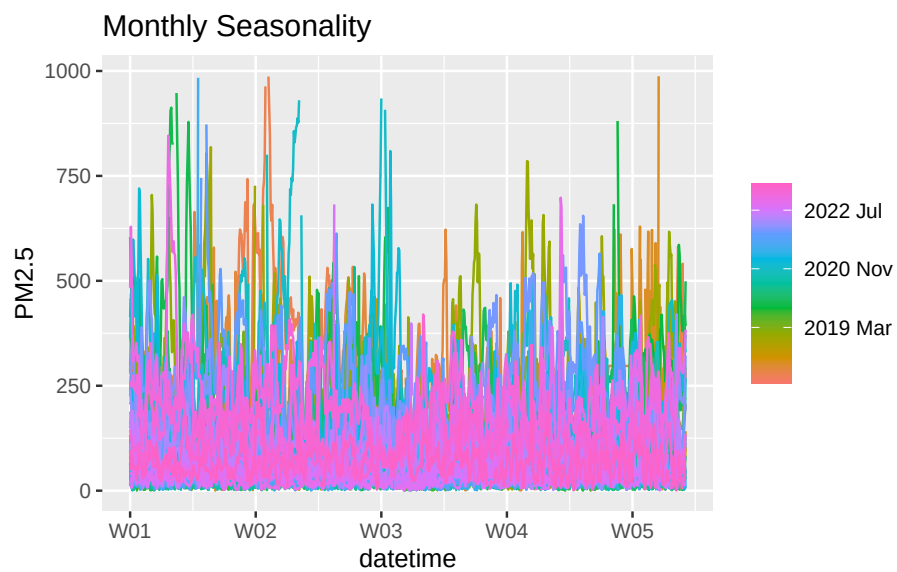
The World Health Organization recommends PM2.5 levels below $35\mu\text{g}/\text{m}^3$ for a daily average and below $12\mu\text{g}/\text{m}^3$ for a yearly average. The daily and yearly recommended levels are plotted in red and blue respectively. Only during the summer months are the PM2.5 levels below the WHO recommended daily limit.

Seasonality

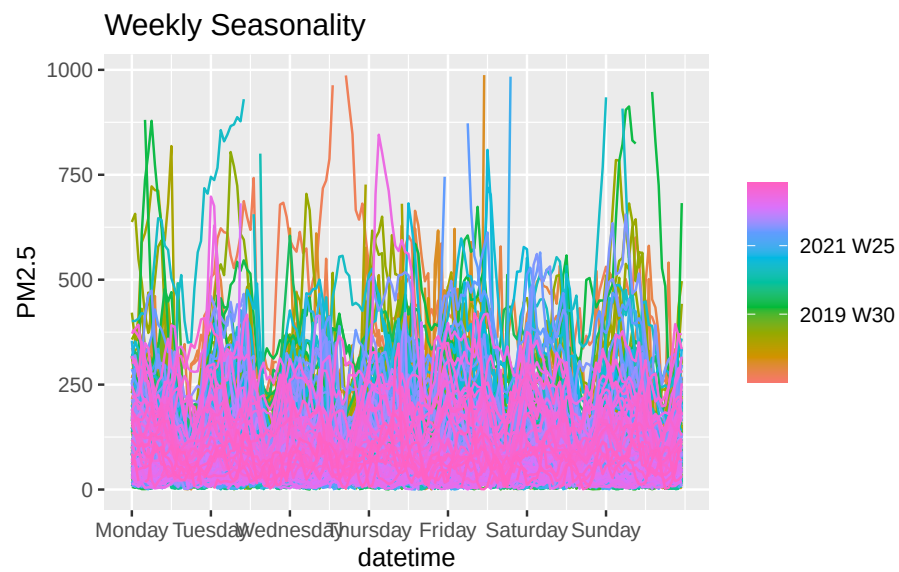
```
gg_season(df, PM2.5, period = "1 year") +  
  labs(title = "Yearly Seasonality")
```



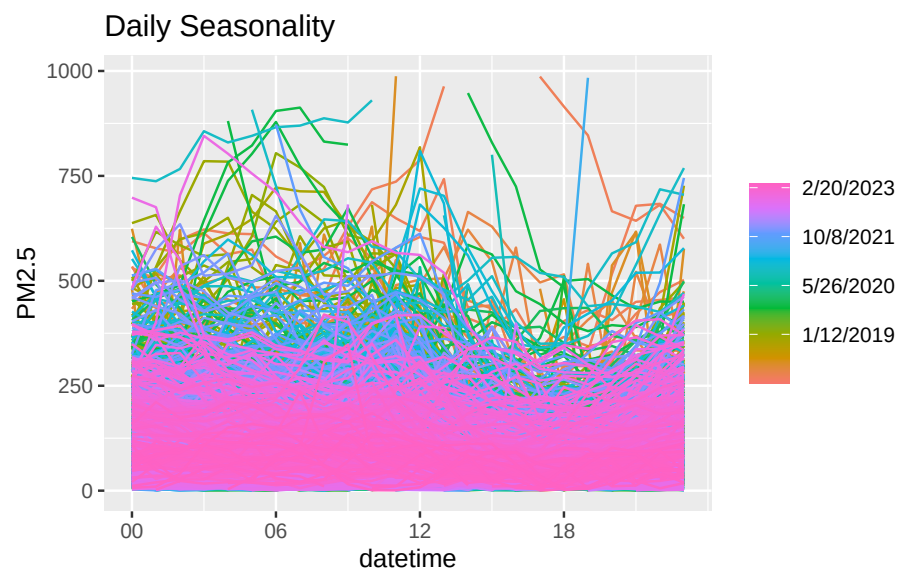
```
gg_season(df, PM2.5, period = "1 month") +  
  labs(title = "Monthly Seasonality")
```



```
gg_season(df, PM2.5, period = "1 week") +  
  labs(title = "Weekly Seasonality")
```

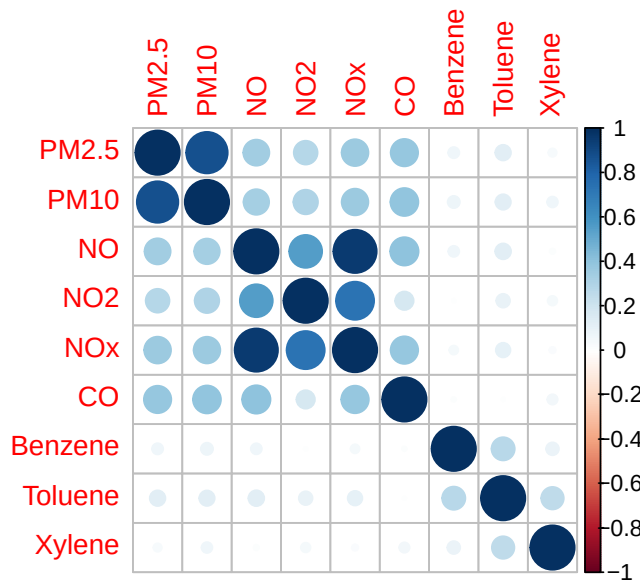


```
gg_season(df, PM2.5, period = "1 day") +  
  labs(title = "Daily Seasonality")
```



Parameter Correlations

```
M <- cor(x = df[, c(2:10)],
        y = df[, c(2:10)],
        use = "na.or.complete")
corrplot(M)
```



High concentrations of CO, NO, NO₂, NO_x, PM_{2.5}, and PM₁₀ generally occur at the same time. High concentrations of Benzene, Toluene, and Xylene generally occur at the same time.

PM_{2.5} as a Stationary Time Series

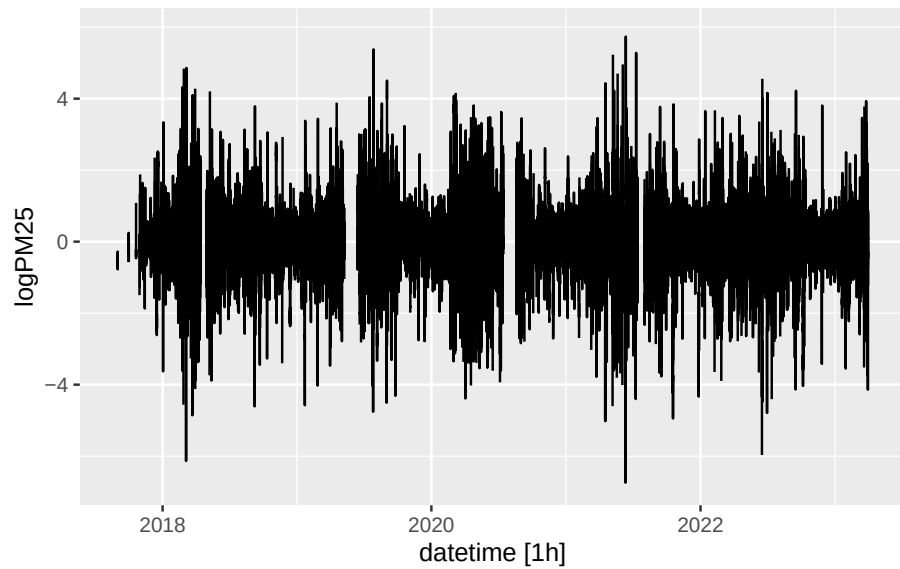
```
df$logPM25 <- log(df$PM2.5) %>%
  difference(24)

# Box-Cox Transformation
BoxCox.lambda(df$PM2.5)

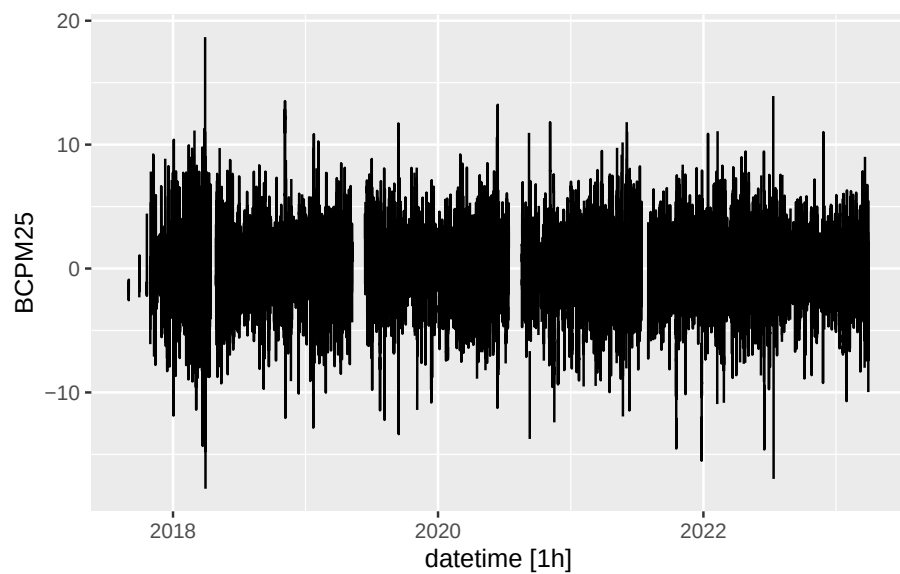
[1] 0.2959454

df$BCPM25 <- BoxCox(df$PM2.5, BoxCox.lambda(df$PM2.5)) %>%
  difference(24)

df %>% autoplot(logPM25)
```



```
df %>% autoplot(BCPM25)
```



Differencing PM2.5 centers the data, and using a log transform centers the data, giving a stationary time series. However, using a Box-Cox transform creates a more stationary time series than the log transform. This method of transformation is likely the better option if we were to move forward with a model choice that requires stationarity.

Check for Constant Variance

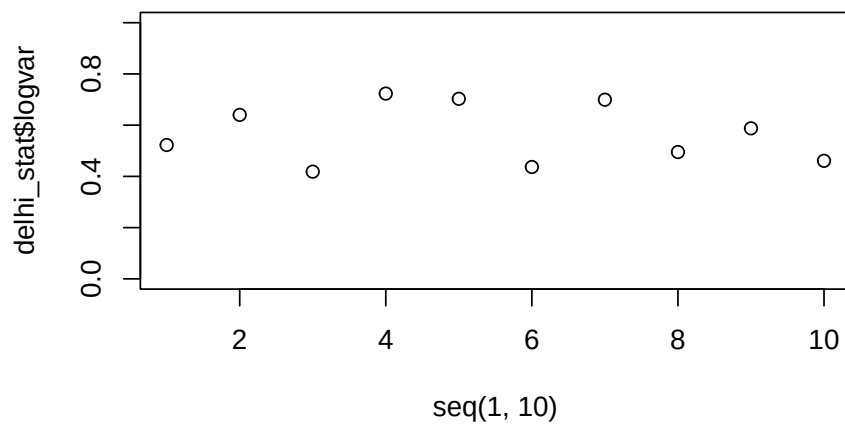
```
delhi_stat <- data.frame(logvar = numeric(10),
                        boxvar = numeric(10))
for (i in 1:10){
  delhi_stat[i,1] <- var(df$logPM25[
    floor(nrow(df)*0.1*(i-1)):floor(nrow(df)*0.1*i)
  ], na.rm = TRUE)

  delhi_stat[i,2] <- var(df$BCPM25[
    floor(nrow(df)*0.1*(i-1)):floor(nrow(df)*0.1*i)
  ], na.rm = TRUE)
}

delhi_stat
```

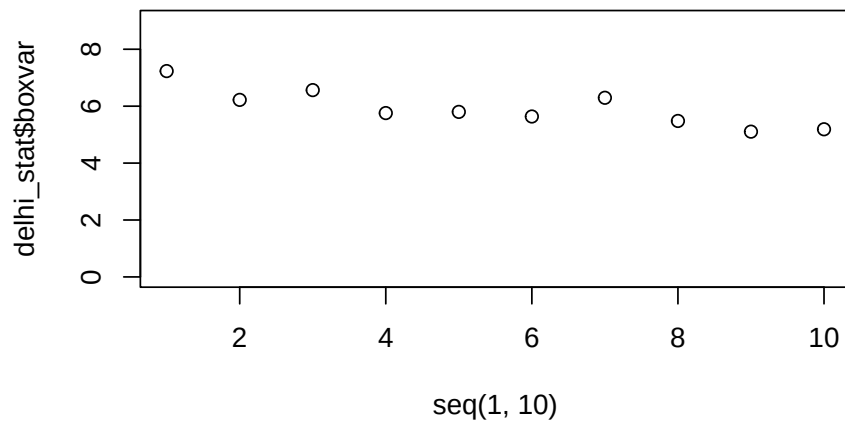
	logvar	boxvar
1	0.5225954	7.232752
2	0.6401363	6.220892
3	0.4184593	6.564328
4	0.7233174	5.757312
5	0.7029033	5.798939
6	0.4367190	5.636711
7	0.6993186	6.294659
8	0.4949893	5.480331
9	0.5876888	5.103950
10	0.4608093	5.187683

```
plot(seq(1,10),delhi_stat$logvar, ylim = c(0,1))
```



The variance changes without transformation.

```
plot(seq(1,10),delhi_stat$boxvar, ylim = c(0,9))
```



The Box-Cox transformation results in stable variances.

World Health Organization Recommended Limits:

CO: 9 ppm
PM_{2.5}: 15 µg/m³
PM₁₀: 45 µg/m³
NO₂: 25 µg/m³
Benzene: As low as possible (<5 µg/m³)
Toluene: 260 µg/m³
Xylene: 870 µg/m³

Scaled Parameters by Recommended Limits

```
df_scaled <- df %>%  
  select(-NOx, -NO) %>%  
  mutate(  
    PM2.5 = PM2.5 / 15,  
    PM10 = PM10 / 45,  
    CO = CO / 9,  
    NO2 = NO2 / 25,  
    Benzene = Benzene / 1,  
    Toluene = Toluene / 260,  
    Xylene = Xylene / 870  
  ) %>%  
  pivot_longer(c(2:8))  
  
ggplot(data = df_scaled) +  
  aes(x = name, y = value, color = name) +  
  geom_boxplot() +  
  scale_y_log10(labels = label_number()) +  
  theme(axis.text.x = element_text(angle = 45, vjust = 0.6)) +  
  labs(  
    x = NULL,  
    y = "Multiple of Recommended Limit",  
    title = "Air Quality Parameters by Recommended Limit"  
  )
```



Conclusion

TBD